NOTE: The document identifier and heading has been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time the document is revised.

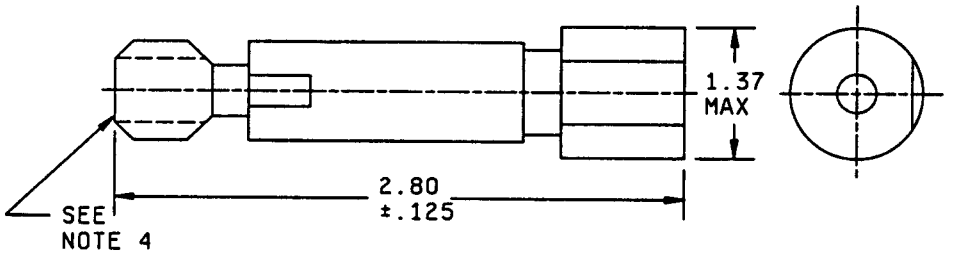
MIL-PRF-39012/120A
9 December 1986
SUPERSEDING
MIL-C-39012/120(EC)
3 October 1980

PERFORMANCE SPECIFICATION SHEET

CONNECTORS, PLUG, ELECTRICAL, COAXIAL, RADIO FREQUENCY, SERIES N (SOCKET CONTACT, SEMIRIGID CABLE, CLASS 2) (FOR USE WITH MIL-C-28830/4)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the connector described herein shall consist of this specification and the latest issue of MIL-PRF-39012.

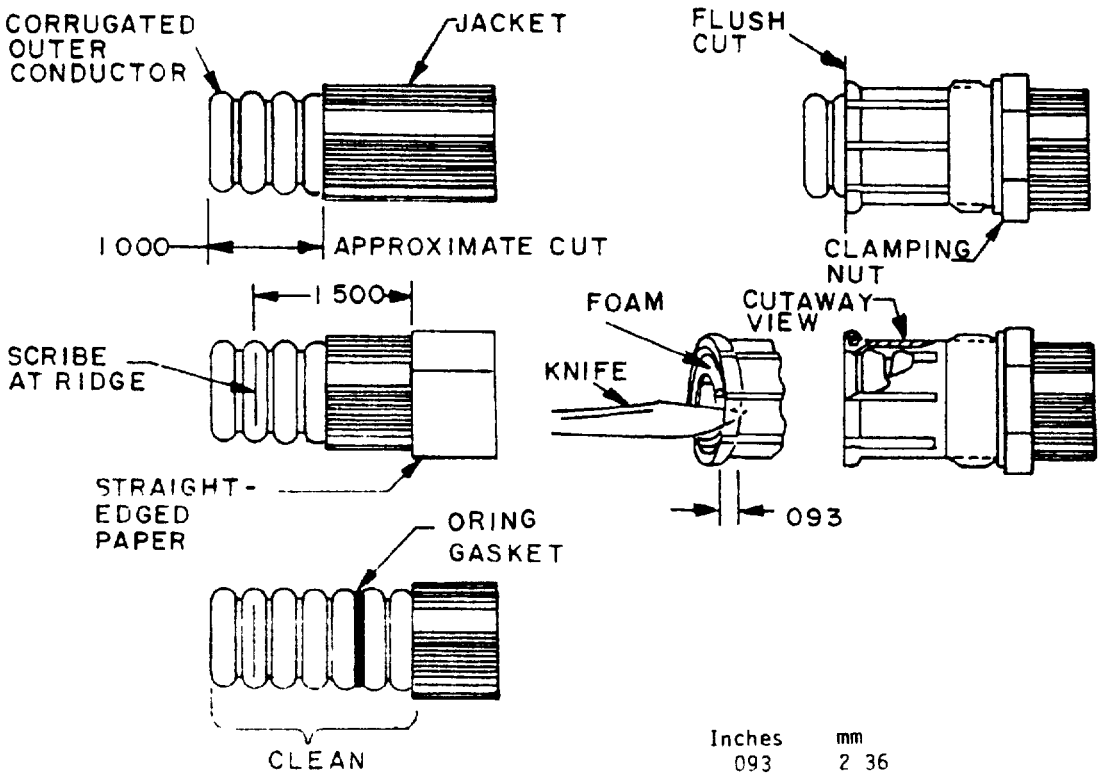


Inches	mm
.125	3.18
1.37	34.8
2.80	71.1

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. All undimensioned pictorial configurations are for reference purposes only.
4. Series N, socket contact interface per MIL-STD-348.

FIGURE 1. General configuration.

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Inches	mm
093	2 36
1 000	25 40
1 500	38.10

NOTES:
1. Dimensions are in inches
2. Metric equivalents are given for general information only
3. Cable outer conductor to be cut at crest only.

FIGURE 2. Cable stripping dimensions for connectors

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ENGINEERING DATA:

Nominal impedance 50 ohms.

Frequency range: 0 to 4.8 GHz.

Voltage rating: 500 volts.

Temperature range: -40°C to +100°C.

Peak RF power: 5.0 kW.

REQUIREMENTS.

Dimensions and configuration. See figure 1.

Force to engage and disengage:

Longitudinal force: Not applicable.

Torque: 6 inch pounds, maximum.

Coupling proof torque: 15 inch pounds, minimum.

Inspection conditions: 6 - 10 inch pounds.

Mating characteristics: See MIL-STD-348 for dimensions.

Hermetical seal: Not applicable.

Leakage (pressurized connectors): Not applicable.

Insulation resistance: Method 302, test condition B, MIL-STD-202.

5,000 megohms, minimum.

Center contact retention: 6 pounds, minimum.

Radial torque: Not applicable.

Corrosion (salt spray). Method 1001, test condition C, in accordance with MIL-STD-1344.

Voltage standing wave ratio (VSWR) per connector. 0.5 to 4.0 GHz-1.2:1.

Swept frequency VSWR test setup:

Item 6. VSWR shall be less than 1.008 +.002 f (f in GHz).

Item 16: VSWR shall be less than 1.008 +.002 f (f in GHz).

Second step of VSWR checkout procedure: VSWR shall be less than 1.012 +.005 f (f in GHz).

Group B inspection: 1.024 +.007.

Qualification and group C inspection: Shall not exceed 1.08.

Connector durability: A minimum of 500 cycles at 12 cycles per minute, maximum.

Contact resistance: In milliohms, maximum.

	Inital	After environment
Inner contact:	2.0	2.0
Outer contact:	0.3	0.3

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Dielectric withstanding voltage at sea level: 5,000 volts, method 301 of MIL-STD-202.

Vibration, high frequency: Method 204, test condition D, MIL-STD-202. Cable clamps shall be mounted on the vibration table 6 inches minimum from the mating face of the connector pair.

Shock: Method 213, test condition I, MIL-STD-202.

Thermal shock: Method 107 of MIL-STD-202 test condition B, except test high temperature shall be +115°C.

Moisture resistance: Method 106 of MIL-STD-202.

No measurements at high humidity. Insulation resistance shall be of at least 200 megohms within 5 minutes after removal from humidity.

Corona level: Not applicable.

RF high potential withstanding voltage: 1,500 volts rms.

Frequency: 5 - 7 MHz.

Leakage current: Not applicable.

Cable retention force: 200 pounds, minimum.

Torque: 96 inch ounces.

Coupling mechanism retention force: 100 pounds, minimum.

RF leakage: -90 dB tested at a frequency of 2 - 3 GHz.

RF insertion loss: dB maximum - 0.10 dB maximum at 4.0 GHz.

Part number: M39012/120-0001.

Revision letters are not used to denote changes due to the extensiveness of the changes.

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy - SH
Air Force: 11, 15, 99

User activities:
Army - AM, AT
Navy - AS, MC, OS
Air Force - 90

Preparing activity:
Navy - EC

Agent:
DLA - ES

(Project 5935-3568-03)

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